

Quiz 4

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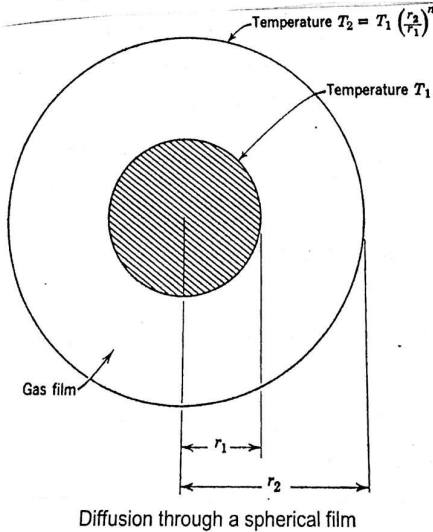
Describe the diffusion in a non-isothermal film in which the temperature changes with the distance according to:

$$\left(\frac{T}{T_1}\right) = \left(\frac{r}{r_1}\right)^n$$

in which T_1 is the temperature at $r = r_1$. Assume as a rough approximation that D_{AB} varies as the $3/2$ - power of the temperature:

$$\left(\frac{D_{AB}}{D_{AB,1}}\right) = \left(\frac{T}{T_1}\right)^{3/2}$$

In which $D_{AB,1}$ is the diffusivity at $T = T_1$. Problems like this arise in connection with drying droplets and diffusion through gas films near spherical catalyst pellets.



$$T = T_1 \left(\frac{r}{r_1}\right)^n \quad D_{AB} = D_{AB,1} \left(\frac{T}{T_1}\right)^{3/2}$$

$$D_{AB} = D_{AB,1} \left[\left(\frac{r}{r_1}\right)^n\right]^{3/2} \quad * C = \frac{P}{RT}$$

$$C D_{AB}(r) = \frac{P}{RT_1} \sqrt{\left(\frac{r}{r_1}\right)^n} D_{AB,1}$$

$$N_A r = X_A N_A r - D_{AB} \frac{dX_A}{dr}$$

$$N_A r (2\pi r^2) \Big|_r - N_A r (2\pi (r+dr)^2) \Big|_{r+dr} = 0$$

$$\frac{d}{dr} (r^2 N_A r) = 0$$

$$\frac{d}{dr} \left(r^2 \frac{C D_{AB}}{1-X_A} \frac{dX_A}{dr} \right) = 0$$

$$\frac{d}{dr} \left(\frac{r^2}{1-X_A} \frac{dX_A}{dr} \right) = 0 \rightarrow \frac{r^2}{1-X_A} \frac{dX_A}{dr} = C_1 \rightarrow \frac{dX_A}{1-X_A} = \frac{C_1}{r^2} dr$$

$$-\ln(1-X_A) = -\frac{C_1}{r} + C_2$$

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$$\text{BC 1: } r=r_1 \quad x_A = x_{A1} \rightarrow -\ln(1-x_{A1}) = -\frac{C_1}{r_1} + C_2$$

$$\text{BC 2: } r=r_2 \quad x_A = x_{A2} \rightarrow -\ln(1-x_{A2}) = -\frac{C_1}{r_2} + C_2$$

$$\ln\left(\frac{1-x_{A2}}{1-x_{A1}}\right) = C_1\left(\frac{1}{r_2} - \frac{1}{r_1}\right) \rightarrow C_1\left(\frac{1}{r_2} - \frac{1}{r_1}\right) r_2$$

$$C_1 = \frac{\ln\left[\frac{(1-x_{A2})}{(1-x_{A1})}\right]}{\frac{1}{r_2} - \frac{1}{r_1}} \quad \therefore C_2 = -\ln(1-x_{A1}) + \frac{1}{r_1} \frac{\ln\left[\frac{(1-x_{A2})}{(1-x_{A1})}\right]}{\left(\frac{1}{r_2} - \frac{1}{r_1}\right)}$$

$$-\ln(1-x_A) = -\frac{1}{r} \frac{\ln\left(\frac{1-x_{A2}}{1-x_{A1}}\right)}{\frac{1}{r_2} - \frac{1}{r_1}} - \ln(1-x_{A1}) + \frac{1}{r_1} \frac{\ln\left(\frac{1-x_{A2}}{1-x_{A1}}\right)}{\frac{1}{r_2} - \frac{1}{r_1}}$$

$$-\ln\left(\frac{1-x_A}{1-x_{A1}}\right) = \frac{-\ln\left(\frac{1-x_{A2}}{1-x_{A1}}\right)}{\frac{1}{r_2} - \frac{1}{r_1}} \left(\frac{1}{r} - \frac{1}{r_1}\right) = -\left[\frac{\frac{1}{r_1} - \frac{1}{r}}{\frac{1}{r_1} - \frac{1}{r_2}}\right] \ln\left[\frac{1-x_{A2}}{1-x_{A1}}\right]$$

$$\frac{1-x_A}{1-x_{A1}} = \left(\frac{1-x_{A2}}{1-x_{A1}}\right)^{\left(\frac{1}{r_1} - \frac{1}{r}\right) / \left(\frac{1}{r_1} - \frac{1}{r_2}\right)}$$

$$N_{Ar} \Big|_{r=r_1} = \frac{C D_{AB}}{1-x_A} \frac{dx_A}{dr} \Big|_{r=r_1} = C D_{AB} \ln\left(\frac{1-x_{A2}}{1-x_{A1}}\right) \frac{1}{r_1^2} \left(\frac{1}{\frac{1}{r_1} - \frac{1}{r_2}}\right)$$

$$W_A = 4\pi r_1^2 N_{Ar} \Big|_{r=r_1} = \frac{4\pi C D_{AB}}{\frac{1}{r_1} - \frac{1}{r_2}} \ln\left(\frac{1-x_{A2}}{1-x_{A1}}\right) \quad \times \text{ almost!}$$